

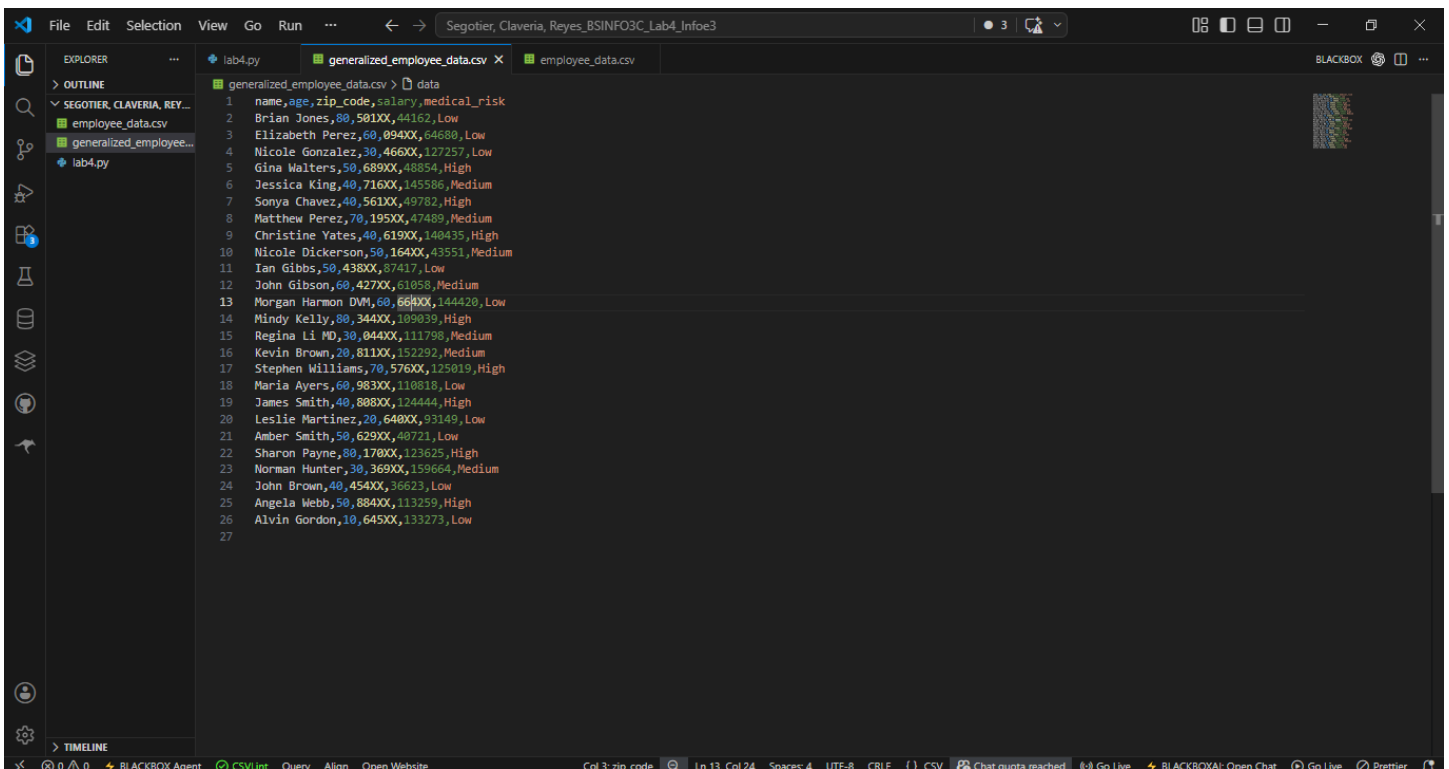
INFOE 3: LAB 4_ Generalization

Joy Melisa S. Segotier

Jorlyn Shane M. Claveria

Maria Theresa N. Reyes

BS INFO 3C



```
1 name,age,zip_code,salary,medical_risk
2 Brian Jones,80,501XX,44162,Low
3 Elizabeth Perez,60,094XX,64680,Low
4 Nicole Gonzalez,30,466XX,127257,Low
5 Gina Walters,50,689XX,48854,High
6 Jessica King,40,716XX,145586,Medium
7 Sonya Chavez,40,561XX,49782,High
8 Matthew Perez,70,195XX,47489,Medium
9 Christine Yates,40,619XX,140435,High
10 Nicole Dickenson,50,164XX,43551,Medium
11 Ian Gibbs,50,438XX,97417,Low
12 John Gibson,60,427XX,61058,Medium
13 Morgan Harmon DVM,60,664XX,144420,Low
14 Mindy Kelly,80,344XX,109039,High
15 Regina Li MD,30,044XX,111798,Medium
16 Kevin Brown,20,811XX,152292,Medium
17 Stephen Williams,70,576XX,125019,High
18 Maria Ayers,60,983XX,110818,Low
19 James Smith,40,888XX,124444,High
20 Leslie Martinez,20,640XX,93149,Low
21 Amber Smith,50,629XX,40721,Low
22 Sharon Payne,80,170XX,123625,High
23 Norman Hunter,30,369XX,159664,Medium
24 John Brown,40,454XX,36623,Low
25 Angela Webb,50,884XX,113259,High
26 Alvin Gordon,10,645XX,133273,Low
27
```

This image displays the **generalized dataset** where precise ages have been rounded down to the nearest decade and zip codes have been masked to prevent re-identification.

The screenshot shows a VS Code editor with a Python script named `lab4.py`. The script performs the following operations:

- Generalizes age by dividing it by 10 and converting it to a string.
- Generalizes zip code by keeping the first three digits and appending "XX".
- Prints the data before and after the transformations.
- Calculates the average salary.

```
17 # Generalize age
18 for record in new_data:
19     age = int(record["age"])
20     record["age"] = str((age // 10) * 10)
21
22
23
24 # Generalize zip code
25 for record in new_data:
26     zip_code = record["zip_code"]
27     record["zip_code"] = zip_code[:3] + "XX"
28
29 # Print BEFORE and AFTER
30 print("BEFORE:")
31 print(raw_data[0])
32
33 print("\nAFTER:")
34 print(new_data[0])
35
36 # Verify average salary
37 total_salary = 0
38
39 for record in new_data:
40     total_salary += int(record["salary"])
41
```

The terminal output shows the results of the script execution:

```
Generalized CSV file created successfully.
PS C:\Users\HYUN GADGETS\Desktop\Segotier, Claveria, Reyes_BSINFO3C_Lab4_Info3> "C:/Users/HYUN GADGETS/Desktop/Segotier, Claveria, Reyes_BSINFO3C_Lab4_Info3/lab4.py"
BEFORE:
{'name': 'Brian Jones', 'age': '80', 'zip_code': '50131', 'salary': '44162', 'medical_risk': 'Low'}
AFTER:
{'name': 'Brian Jones', 'age': '80', 'zip_code': '501XX', 'salary': '44162', 'medical_risk': 'Low'}
Average Salary: 97536.6
Generalized CSV file created successfully.
```

This image demonstrates the generalization of zip codes by using string slicing to keep the first three digits and appending "XX" to hide the last two.

The screenshot shows a VS Code editor with a Python script named `lab4.py`. The script performs the following operations:

- Reads data from a CSV file using `csv.DictReader`.
- Creates a deep copy of the data.
- Categorizes age into life stages: "Under 30", "30 to 50", and "Over 50".
- Generalizes zip code by keeping the first three digits and appending "XX".
- Prints the data before and after the transformations.
- Calculates the average salary.

```
10 reader = csv.DictReader(file)
11 for row in reader:
12     raw_data.append(row)
13
14 # Create a deep copy of the data
15 new_data = copy.deepcopy(raw_data)
16
17 # Categorize age into life stages
18 for record in new_data:
19     age = int(record["age"])
20
21     if age < 30:
22         record["age"] = "Under 30"
23
24     elif age <= 50:
25         record["age"] = "30 to 50"
26
27     else:
28         record["age"] = "Over 50"
29
30
31
32 # Generalize zip code
33 for record in new_data:
34     zip_code = record["zip_code"]
35     record["zip_code"] = zip_code[:3] + "XX"
36
37 # Print BEFORE and AFTER
38 print("BEFORE:")
39 print(raw_data[0])
40
41 print("\nAFTER:")
42 print(new_data[0])
43
44 # Verify average salary
45 total_salary = 0
46
47 for record in new_data:
48     total_salary += int(record["salary"])
49
```

The terminal output shows the results of the script execution:

```
Generalized CSV file created successfully.
PS C:\Users\HYUN GADGETS\Desktop\Segotier, Claveria, Reyes_BSINFO3C_Lab4_Info3> "C:/Users/HYUN GADGETS/Desktop/Segotier, Claveria, Reyes_BSINFO3C_Lab4_Info3/lab4.py"
BEFORE:
{'name': 'Brian Jones', 'age': '80', 'zip_code': '50131', 'salary': '44162', 'medical_risk': 'Low'}
AFTER:
{'name': 'Brian Jones', 'age': 'Over 50', 'zip_code': '501XX', 'salary': '44162', 'medical_risk': 'Low'}
Average Salary: 97536.6
Generalized CSV file created successfully.
```

This image shows the implementation of the "Extra" task, which involves using conditional logic to categorize ages into specific "Life Stages" like "Under 30," "30 to 50," and "Over 50".